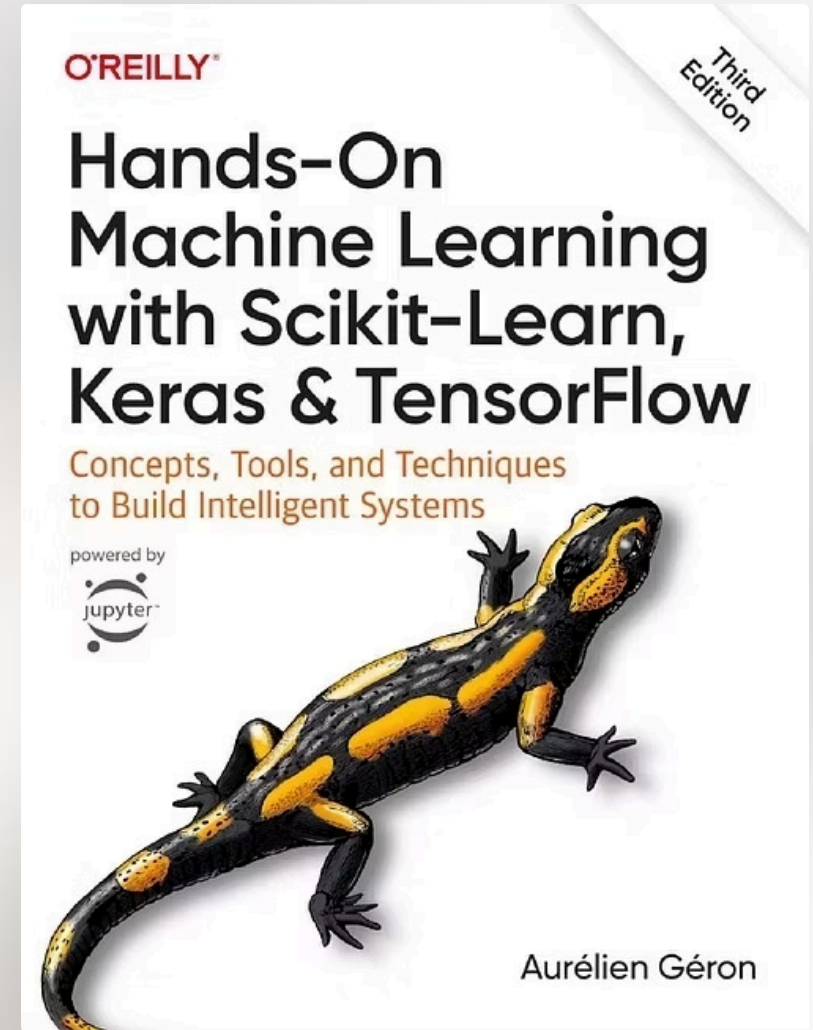


Machine Learning

Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

RP by Rodrigo Martins Pagliares

Last update: 08/07/2025



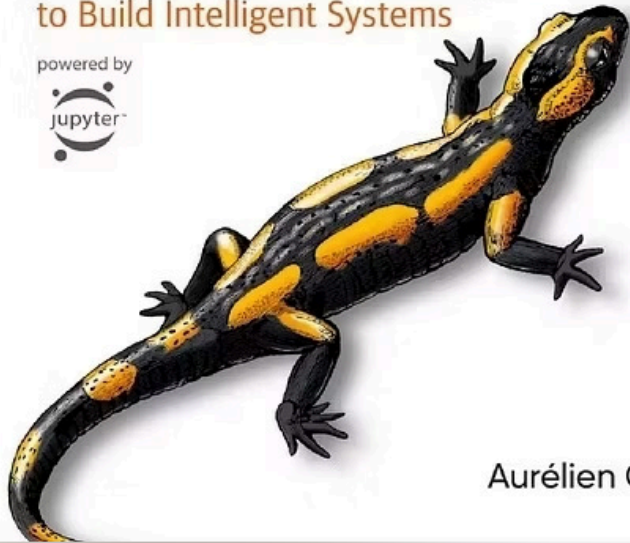
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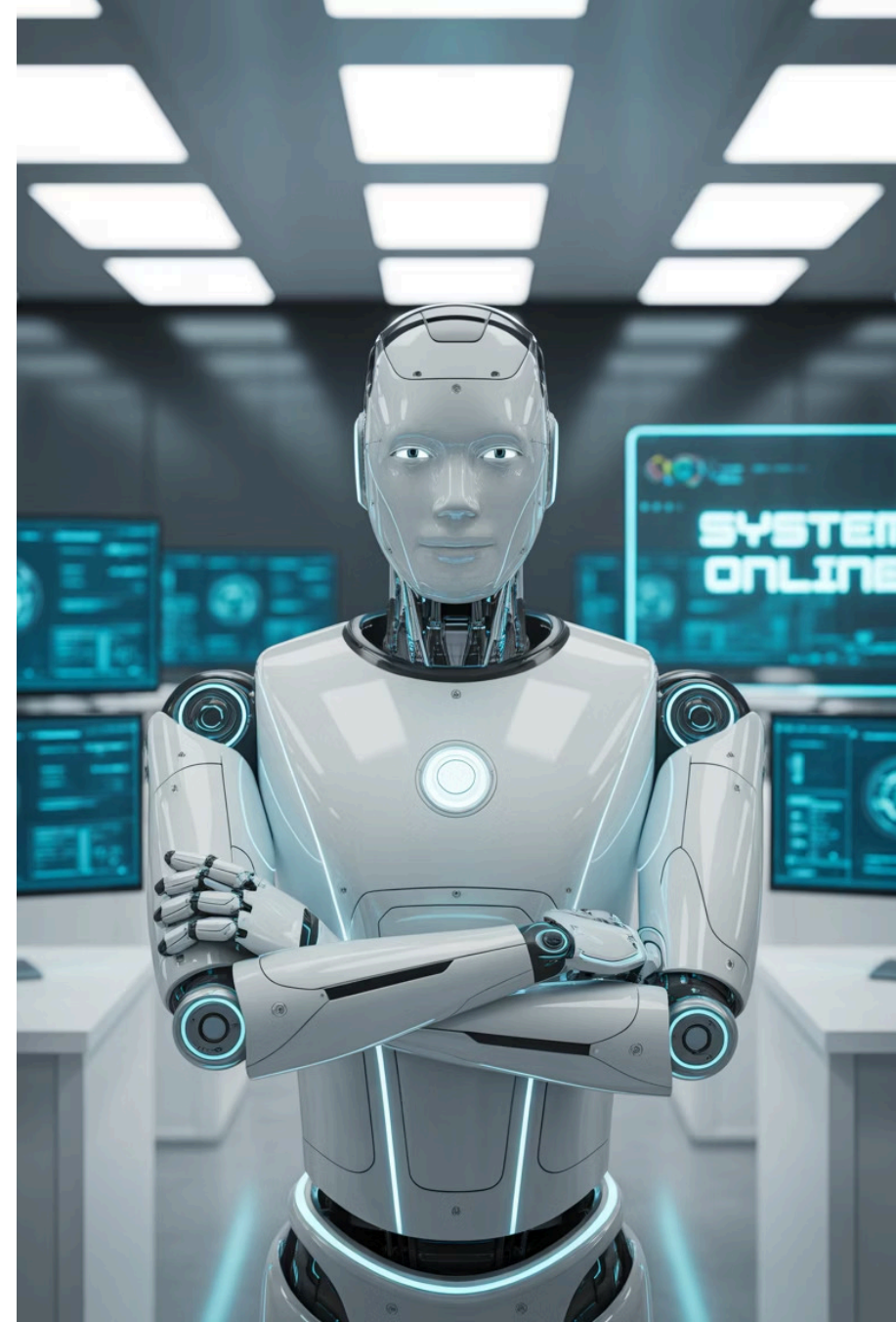
Aurélien Géron

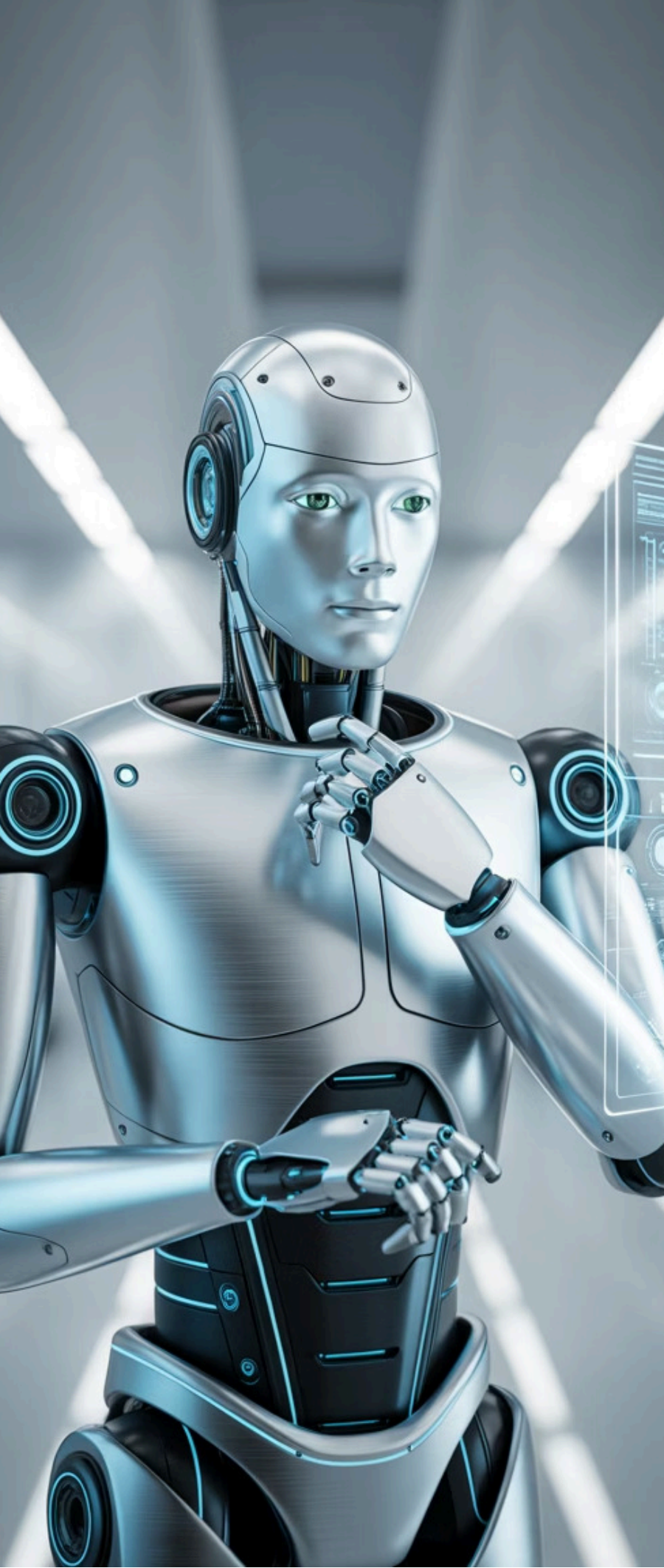
Machine Learning

Chapter 1 – The Machine Learning Landscape

Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
5. **Main Challenges of Machine Learning**
6. Testing and Validating



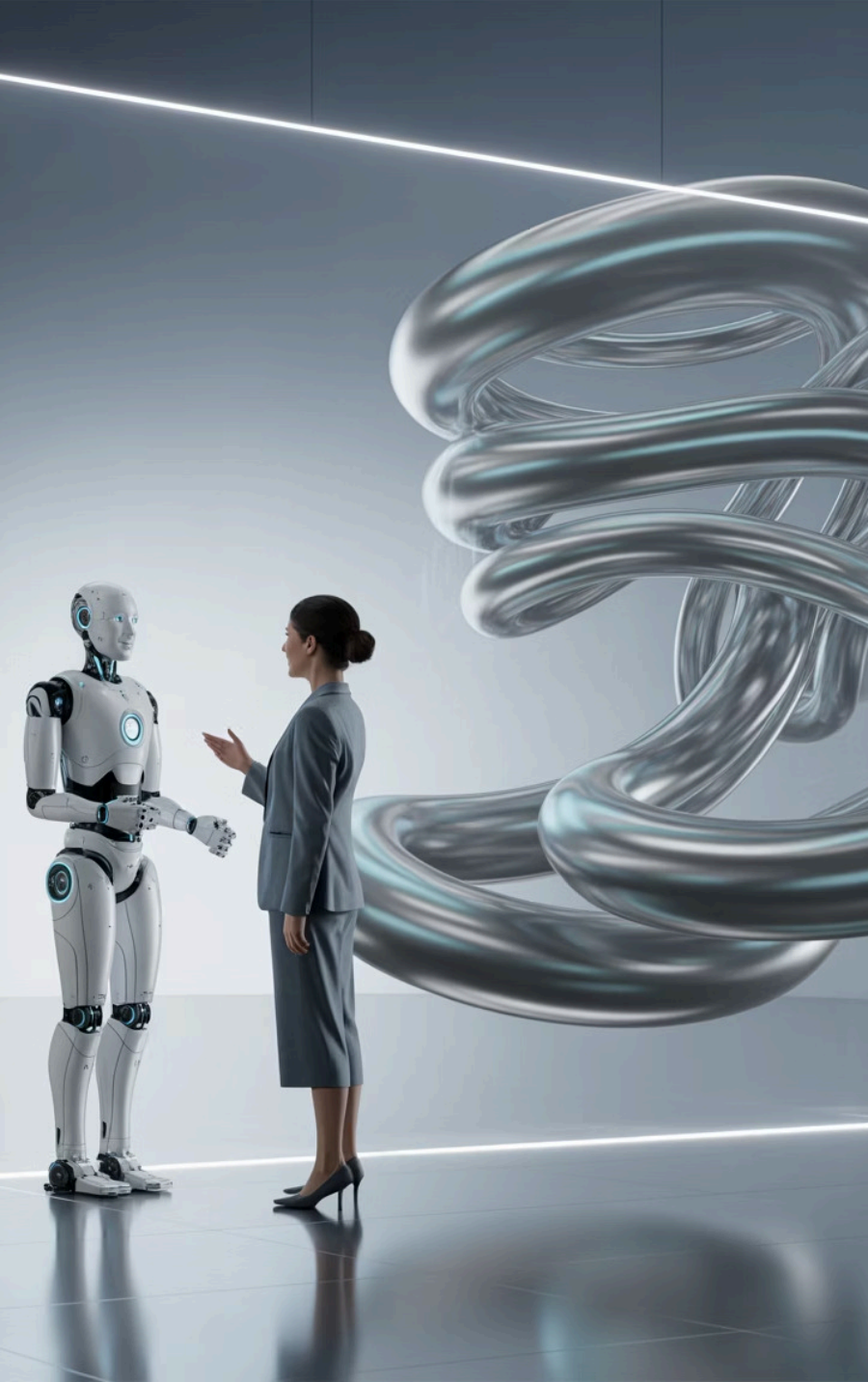


Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. Types of Machine Learning Systems
 - Classification criteria #1: Supervision during training
 - Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning
 - Classification criteria #2: Approach to Learning Method
 - Batch and Online (Incremental) Learning
 - Classification criteria #3: Learning Method
 - Instance-Based and Model-Based Learning
5. Main Challenges of Machine Learning
 - Insufficient Quantity of Training Data
 - Non-representative Training Data
 - Poor-Quality Data
 - Irrelevant Features
 - Overfitting the Training Data
 - **Underfitting the Training Data**
 - Stepping Back: Reviewing the Chapter So Far
6. Testing and Validating
 - Hyperparameter Tuning and Model Selection
 - Data Mismatch

Main Challenges of Machine Learning

Insufficient Quantity of Training Data, Non-representative Training Data, Poor-Quality Data, Irrelevant Features, **Overfitting the Training Data**, **Underfitting the Training Data**, Stepping Back: Reviewing the Chapter So Far



Underfitting the Training Data

Definition of Underfitting

- Occurs when the **model is too simple** to learn the data's underlying **patterns**
- Leads to **poor performance** on both **training** and **test sets**
- Example:** Linear model of life satisfaction might **underfit** if the **data is more complex**

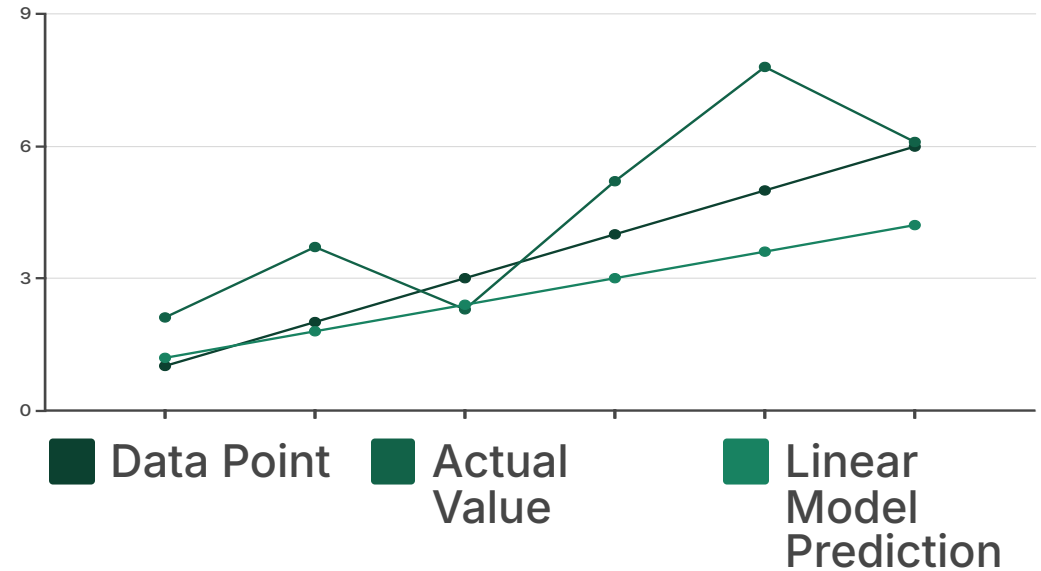
Example of Underfitting

Case Study: Life Satisfaction Model

Linear model may be too simplistic for complex data

Results: Model predictions remain inaccurate even on training data

Conclusion: A simple model like a linear regression might not capture all the complexities in the data





Solutions to Underfitting

1. Select a More Powerful Model

Choose a more complex model with more parameters

Models with higher complexity can capture more intricate patterns in the data

Examples:

- Decision trees
- Support vector machines
- Neural networks

Solutions to Underfitting (contd.)

2. Feature Engineering



Add more relevant features

Include additional variables that might help explain the target variable



Transform existing features

Create polynomial features, interaction terms, or other transformations



Example

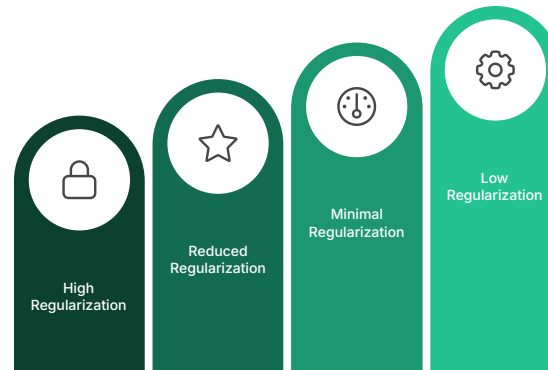
Transform linear features into polynomial features to capture non-linear relationships

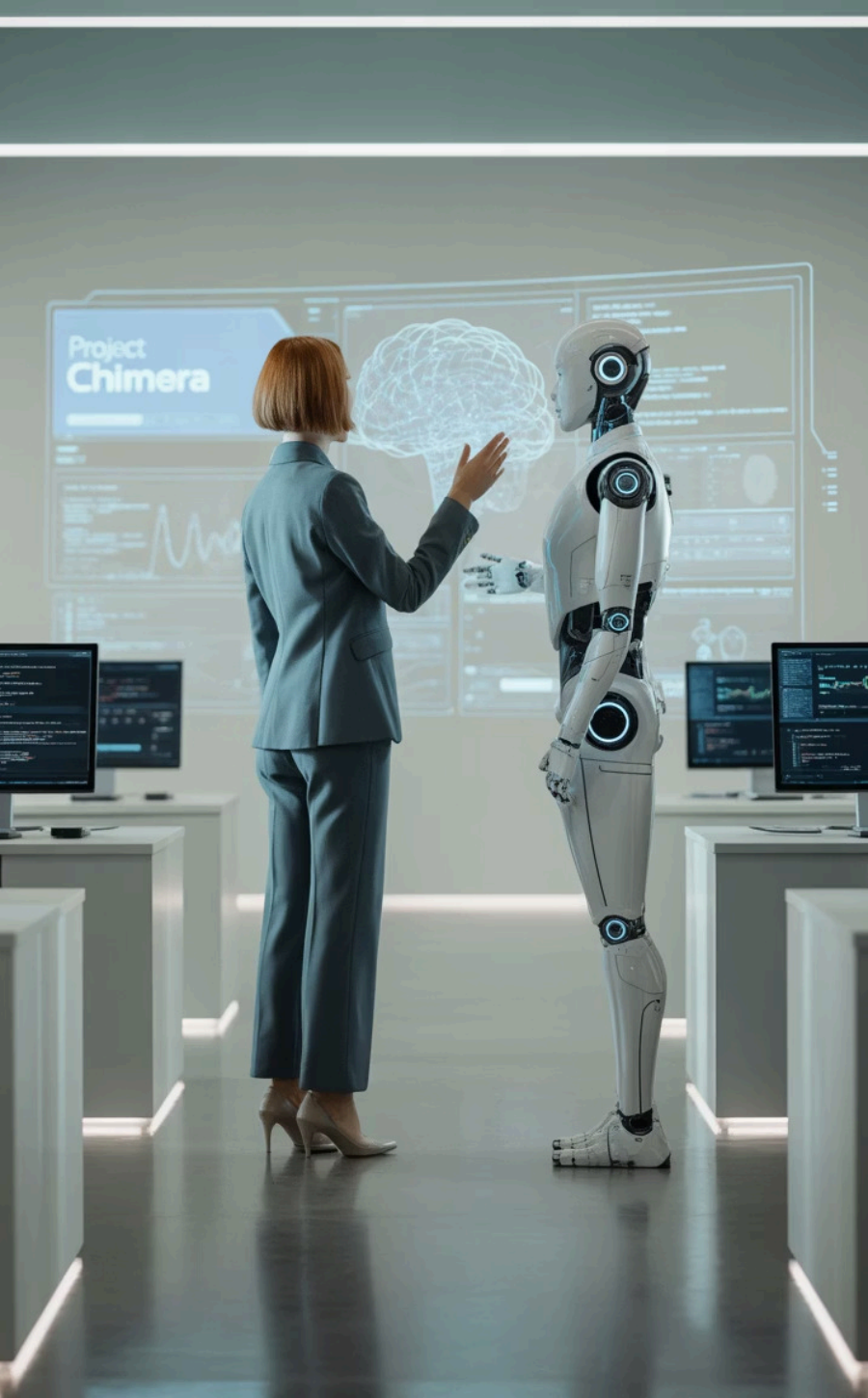
Solutions to Underfitting (contd.)

3. Reduce Model Constraints

Reduce regularization to allow the model to learn more complex patterns

Example: Lower the regularization hyperparameter to give the model more flexibility





Recap

Summary of Underfitting

The Problem

Simple models might not capture the complexity of the data

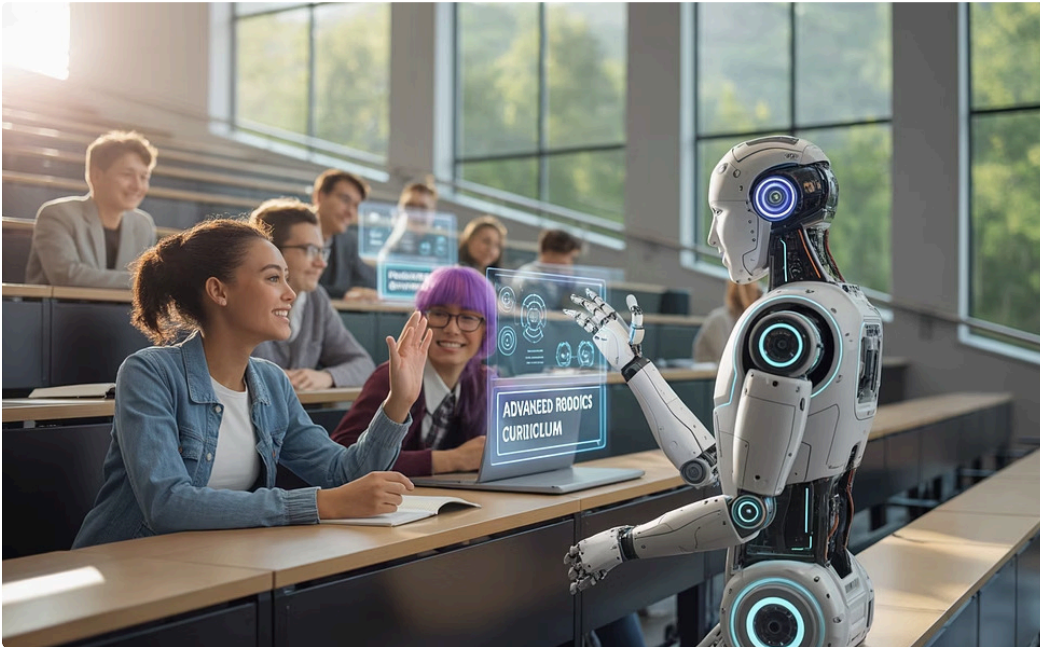
Solutions

- Use a more powerful model
- Improve features
- Reduce regularization constraints

Next: Stepping Back: Reviewing the Chapter So Far

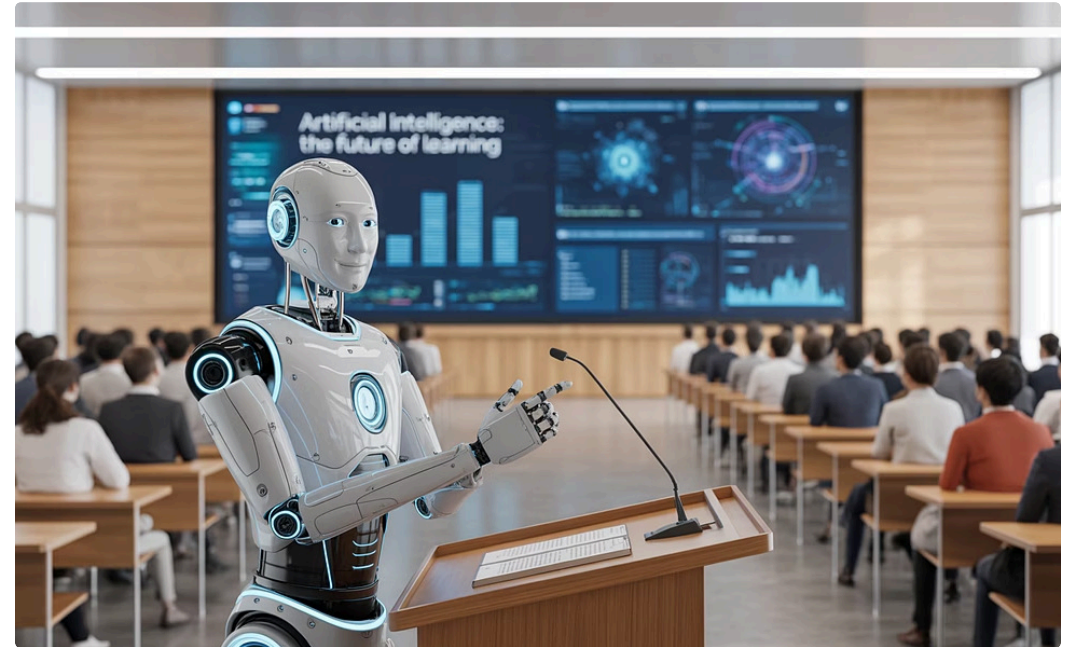


Q&A



Open Floor

Open floor for discussion or clarifications



Discussion

Share your thoughts and questions about AI concepts